Т3

$$p(x) = \begin{cases} e^{-\frac{x}{\theta}}, & x \geq 0 \\ 0, & x < 0 \end{cases} \qquad n = 3$$

$\tilde{\theta}$: $\tilde{\theta}_1 = \overline{x} = \frac{1}{3}\sum_{i=1}^{3} x_i$

$\tilde{\theta}_3 = x_{(2)}$ ~ 2ая порядковая статистика

а)

$M\tilde{\theta}_1 = M\frac{1}{n}\sum_{i=0}^{n} x_i = \frac{1}{n}\sum Mx_i = M\theta$ – несмещ

б) $F_{x(k)} = \frac{n!}{(k-1)!(n-k)!} \cdot \int_0^{F(x)} t^{k-1}(1-t)^{n-k}\,dt$

$$\rho_{x(k)} = \frac{n!}{(k-1)!(n-k)!}\left(F^{k-1}(x)\,(1-F(x))^{n-k}\right)\rho(x)$$

$\cancel{Mx_{(2)} = \int_0^{\infty} x\rho_{x(2)}\,dx = \frac{6}{\theta}\int_0^{\infty} x}$ $\quad F(x) = \int_{-\infty}^{\infty}\rho(x)\,dx = \int_0^{\infty}\frac{1}{\theta}e^{-\frac{x}{\theta}}\,dx =$

$= 1 - e^{-x/\theta}, x \geq 0$

$\rho_{x(2)} = 6\,(1-e^{-x/\theta})\,e^{-x/\theta}\cdot\frac{1}{\theta}e^{-x/\theta}$

$\rho_{x(2)} = \frac{6}{\theta}(1-e^{-x/\theta})\cdot e^{-\frac{2x}{\theta}}, x \geq 0$

$Mx_{(2)} = \frac{6}{\theta}\int_0^{\infty} x\,(1-e^{-x/\theta})\,e^{-\frac{2x}{\theta}}\,dx =$

$= \frac{6}{\theta}\int_0^{\infty} x e^{-2x/\theta}\,dx - \frac{6}{\theta}\int_0^{\infty} x e^{-x/\theta}e^{-2x/\theta}\,dx =$

$= \frac{6}{\theta}\int_0^{\infty} x e^{-2x/\theta}\,dx - \frac{6}{\theta}\int_0^{\infty} x e^{-\frac{3x}{\theta}}\,dx = \frac{6}{\theta}\cdot\frac{\theta^2}{4} - \frac{6}{\theta}\frac{\theta^2}{9} =$

$= \frac{5}{6}\theta \Rightarrow \tilde{\theta}_3' = \frac{6}{5}\tilde{\theta}_3, \quad M\tilde{\theta}_3' = \frac{6}{5}\cdot\frac{5}{6}M\theta = M\theta$

~ смещ. ~ несмещ.

b)

$$\cdot\ D\tilde{\Theta}_1 = \frac{1}{9}\cdot 3D\Theta = \frac{1}{3}\Theta^2$$

$$D\tilde{\Theta}_3'$$

$$M\tilde{\Theta}_3^2 = \int_0^\infty x^2 \frac{6}{\Theta}(1-e^{-x/\Theta})e^{-2x/\Theta}\,dx =$$

$$= \frac{6}{\Theta}\int_0^\infty x^2 e^{-2x/\Theta} - \frac{6}{\Theta}\int_0^\infty x^2 e^{-3x/\Theta}\,dx =$$

$$= \frac{6}{\Theta}\cdot\frac{\Theta^3}{4} - \frac{6}{\Theta}\cdot\frac{2\Theta^3}{27} = \frac{27\Theta^2}{18} - \frac{8\Theta^2}{18} = \frac{19}{18}\Theta^2$$

$$D\tilde{\Theta}_3 = \frac{19}{18}\Theta^2 - \frac{25}{36}\Theta^2 = \frac{38}{36}\Theta^2 - \frac{25}{36}\Theta^2 = \frac{13}{36}\Theta^2$$

$$D\tilde{\Theta}_3' = \frac{36}{25}\cdot D\tilde{\Theta}_3 = \frac{36}{25}\cdot\frac{13}{36}\Theta^2 = \frac{13}{25}\Theta^2$$

$$D\tilde{\Theta}_3' > D\tilde{\Theta}_1$$

$$\frac{13}{25}\Theta^2 > \frac{1}{3}\Theta^2 \implies \tilde{\Theta}_1 \text{ - более эфф. оценка}$$

$$\frac{39}{75}\Theta^2 > \frac{25}{75}\Theta^2$$

c)

Неизм модели

$$\int_{-\infty}^{\infty} dF(x) = \int_{-\infty}^{\infty} d\frac{\partial F}{\partial \theta} \qquad + \theta \, d\square$$

$$\ln p = \ln e^{-x/\theta} - \ln\theta = -\frac{x}{\theta} - \ln\theta$$

$$\frac{\partial \ln p}{\partial \theta} = \frac{x}{\theta^2} - \frac{1}{\theta} = \frac{x-\theta}{\theta^2}$$

$$I_1(\theta) = \int_{-\infty}^{\infty} \left(\frac{\partial \ln p}{\partial \theta}\right)^2 p\,dx = \int_0^{\infty} \left(\frac{x-\theta}{\theta^2}\right)^2 \frac{e^{-x/\theta}}{\theta}\,dx =$$

$$= \frac{1}{\theta^5}\int_0^{\infty} (x-\theta)^2 e^{-x/\theta}\,dx = -\frac{1}{\theta^4}\int_0^{\infty} (x-\theta)^2\,d e^{-x/\theta} =$$

$$= -\frac{1}{\theta^4}\left((x-\theta)^2 \cdot e^{-x/\theta}\Big|_0^{\infty} - \int_0^{\infty} e^{-x/\theta} \cdot 2(x-\theta)\,dx\right) =$$

$$= -\frac{1}{\theta^4}\left(-\theta^2 + \frac{2}{\theta}\int_0^{\infty} (x-\theta)\,d e^{-x/\theta}\right) =$$

$$= -\frac{1}{\theta^4}\left(-\theta^2 + \frac{2}{\theta}\left((x-\theta)e^{-x/\theta}\Big|_0^{\infty} - \theta\int_0^{\infty} e^{-x/\theta}\,\frac{dx}{\theta}\right)\right) =$$

$$= -\frac{1}{\theta^4}\left(-\theta^2 + \frac{2}{\theta}\left(\theta + \theta e^{-x/\theta}\Big|_0^{\infty}\right)\right) =$$

$$= -\frac{1}{\theta^4}\left(-\theta^2 + 2 - 2\right) = \frac{1}{\theta^2}$$

$I(\theta)$ – непр на $\forall$ компакте из $\square$

$I(\theta) > 0$

$\Longrightarrow$ модель регулярна

Оценки $\tilde{\theta}_1$ и $\tilde{\theta}_3'$ - несмещ.

$D\tilde{\theta}_1$ и $D\tilde{\theta}_3'$ ограничены на ~~θ~~ компакте $\boxed{\theta}$
$\theta > 0$

Оценки регулярные

$$D\tilde{\theta} = \frac{(g'(\theta))^2}{n\, I(\theta)} = \frac{1}{3 \cdot \frac{1}{\theta^2}} = D\tilde{\theta}_1$$

$\Longrightarrow \tilde{\theta}_1$ - эфф. оценка по дост.
(Оптимальная оценка в классе регулярных)

$$D\tilde{\theta}_3' > \frac{1}{3 \cdot \frac{1}{\theta^2}} \Longrightarrow \tilde{\theta}_3' \text{ - эфф. оценка}$$

~~Т4~~

~~D: $0\{p\} + 2\{p\}$; $p+p=2p$~~

~~C: $\int_{-1}^{1} c\,dx = 2c$~~

~~$2c + 2p = 1$~~

~~$p_c = \frac{1}{2}$~~

~~$2c = \frac{1}{2}$~~

~~$c = \frac{1}{4}$; $p = \frac{1}{4}$~~

Т4

Θ - вер-ть попасть в дискр. часть

$P(x=0) = P(x=2) = \frac{\Theta}{2}$

непр. $p_{нов}(x) = \frac{1-\Theta}{2}$ $\qquad \xi \sim p(x) = \frac{1-\Theta}{2}\{(-1,1)\} +$

а) $M\xi = \int_{-1}^{1} x \cdot \frac{1-\Theta}{2}\,dx + 0 \cdot \frac{\Theta}{2} + \frac{\Theta}{2}\{0\} + \frac{\Theta}{2}\{2\}$

$+ 2 \cdot \frac{\Theta}{2} = \Theta = \bar{x}$ ОММ несмещ.

$L(\Theta) = \left(\frac{\Theta}{2}\right)^{n_0} \left(\frac{\Theta}{2}\right)^{n_2} \left(\frac{1-\Theta}{2}\right)^{n_c}$

$L(\Theta) = \left(\frac{\Theta}{2}\right)^{n_d} \left(\frac{1-\Theta}{2}\right)^{n_c}$

$\ln L(\Theta) = n_d \ln \frac{\Theta}{2} + n_c \ln\left(\frac{1-\Theta}{2}\right) =$

$= n_d \ln \Theta + n_c \ln(1-\Theta) - n \ln 2$

$(\ln L(\Theta))' = \frac{n_d}{\Theta} - \frac{n_c}{1-\Theta} = 0$

$n_d(1-\Theta) = n_c \Theta$; $\Theta(\overbrace{n_c + n_d}^{=n}) = n_d \Rightarrow \hat{\Theta} = \frac{n_d}{n_c + n_d}$

$$\tilde{\theta} = \frac{n_0 + n_2}{n}$$

$$\sup_{\theta \in \Theta} L(\theta) \longrightarrow \tilde{\theta} = \frac{n_0 + n_2}{n}$$

n_0 - число 0

n_2 - число 2

ОММП

b) $\theta = \bar{x}$ - несмещ.

$$\tilde{\theta} = \frac{n_0 + n_2}{n}$$

$$X_i = \begin{cases} 1, & x \in D \\ 0, & x \in C \end{cases}$$

$$\tilde{\theta} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

$$\cancel{M\tilde{\theta}} \quad MX_i = 1 \cdot \theta + 0(1-\theta) = \theta$$

$$M\tilde{\theta} = M\frac{1}{n}\sum X_i = \frac{1}{n}\sum MX_i = MX_i = \theta$$

несмещ.

c) $$I(\theta) = M\left[\left(\frac{\partial}{\partial\theta} \ln p\right)^2\right] =$$

$$= \int_{-1}^{1} \left(-\frac{1}{1-\theta}\right)^2 \cdot \frac{1-\theta}{2} dx + \frac{\theta}{2} \cdot \left(\frac{1}{\theta}\right)^2 + \frac{\theta}{2}\left(\frac{1}{\theta}\right)^2 =$$

$$= \int_{-1}^{1} \frac{1}{(1-\theta)2} dx + \frac{1}{2\theta} + \frac{1}{2\theta} =$$

$$= \frac{1}{1-\theta} + \frac{1}{\theta} = \frac{1}{\theta(1-\theta)}$$

$I(\theta)$ - разрывна на $\square$

и не положительна на $\square$

$\Longrightarrow$ модель не регулярна $\Rightarrow$ не можем исп. Неравенство Фишера